

KILSBERGEN-SUTTARBODA, SwedenWMO: **024520**Lat: **59.2996N**Lon: **14.8956E**Elev: **719**StdP: **14.32**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	10.6	1.2	6.0	10.5	5.5	7.5	13.7	23.2	40.4	20.6	38.5	5.9	310	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	77.5	63.0	74.5	61.8	71.4	60.2	65.7	73.0	63.9	70.8	62.2	68.7	5.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.3	89.4	67.8	61.4	83.6	66.5	59.5	77.9	64.7	30.9	72.7	29.5	71.1	28.3	68.8	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.4	13.7	2.8	81.8	5.2	2.9	-1.0	83.9	-4.0	85.6	-6.9	87.2	-10.6	89.3		
(5)			2.1	67.9	4.7	2.0	-1.3	69.3	-4.1	70.5	-6.7	71.7	-10.1	73.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	26.2	26.9	32.4	41.8	50.8	57.4	62.3	60.1	53.0	42.6	34.9	30.2
	DBStd	14.29	8.22	7.51	7.40	6.45	6.94	5.61	5.45	4.76	5.14	6.22	6.83	8.28
	HDD50	3603	737	648	546	258	73	5	0	1	28	240	452	615
	HDD65	7981	1202	1068	1010	695	443	237	123	166	360	695	902	1080
	CDD50	1160	0	0	0	13	96	228	381	314	118	10	0	0
	CDD65	65	0	0	0	0	2	10	39	14	0	0	0	0
	CDH74	304	0	0	0	0	11	55	193	45	0	0	0	0
	CDH80	25	0	0	0	0	0	3	19	3	0	0	0	0
Wind	WSAvg	6.9	7.6	7.1	7.6	7.1	6.7	6.4	6.0	5.6	6.4	6.9	7.5	8.1
Precipitation	PrecAvg	29.4	2.1	1.7	1.4	1.7	2.5	2.5	3.4	3.0	2.6	2.9	2.7	2.2
	PrecMax	40.8	4.8	3.7	2.5	3.4	5.5	4.8	8.3	6.9	5.4	7.2	6.6	4.5
	PrecMin	22.8	0.2	0.7	0.1	0.5	1.3	1.0	0.7	0.3	0.6	0.0	0.9	0.4
	PrecStd	4.0	1.2	0.8	0.6	0.9	1.1	1.2	1.9	1.5	1.4	1.7	1.4	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	44.0	55.2	65.6	75.7	79.7	82.6	78.5	70.1	58.6	51.2	47.7
		MCWB	42.6	40.7	45.6	50.8	59.0	61.8	65.0	63.3	60.2	54.3	49.7	45.4
	2%	DB	41.0	40.9	49.2	61.1	71.0	75.0	78.9	74.4	66.2	55.3	48.6	44.8
		MCWB	39.6	38.5	41.5	48.1	55.7	60.2	64.4	63.2	58.0	52.7	47.0	43.0
	5%	DB	38.9	38.7	45.4	56.6	67.1	71.5	75.9	71.3	63.0	53.3	46.7	42.7
		MCWB	37.9	36.9	39.5	46.6	54.9	58.5	63.1	61.8	57.0	51.2	44.9	40.7
	10%	DB	36.9	36.8	42.1	52.6	62.8	67.8	72.5	68.2	60.4	51.4	44.4	40.6
		MCWB	36.0	35.4	37.9	43.8	52.7	56.7	61.7	60.3	55.6	49.7	43.1	39.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	42.1	45.6	53.3	61.8	64.3	68.8	67.7	62.2	55.7	50.4	46.8
		MCDB	43.6	42.9	53.5	62.9	70.8	72.5	76.3	73.5	67.2	57.2	50.9	47.1
	2%	WB	40.3	39.2	42.4	49.6	59.1	62.1	66.4	65.4	59.8	53.6	47.7	43.8
		MCDB	40.9	40.1	47.3	58.3	66.0	71.5	74.4	71.0	64.5	54.7	48.3	44.4
	5%	WB	38.1	37.4	40.6	47.4	56.3	60.1	64.7	63.3	58.1	51.6	45.5	41.4
		MCDB	38.8	38.5	44.2	55.0	63.7	68.7	72.8	68.0	62.2	52.9	46.4	42.1
	10%	WB	36.2	35.4	38.6	45.0	53.9	58.1	63.1	61.6	56.5	49.8	43.4	39.4
		MCDB	36.9	36.6	41.4	51.2	61.2	65.6	70.4	66.6	59.9	51.2	44.2	40.2
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.3	10.4	13.4	14.3	13.8	13.2	12.1	10.9	8.5	6.4	6.8
		MCDBR	8.0	8.3	15.0	18.5	19.6	18.4	17.3	15.5	13.7	9.3	8.7	7.6
		MCWBR	8.3	6.6	9.4	10.0	10.3	9.1	7.7	7.3	7.7	7.3	8.5	8.0
	5% WB	MCDBR	7.9	7.4	13.4	16.9	16.8	16.5	15.3	13.3	12.5	8.6	8.3	7.9
MCWBR		8.1	6.3	9.2	9.9	10.1	9.1	7.9	7.9	8.0	7.5	8.1	8.4	
Clear-Sky Solar Irradiance	taub		0.267	0.307	0.329	0.366	0.362	0.352	0.374	0.366	0.335	0.325	0.291	0.249
	taud		2.223	2.309	2.372	2.339	2.413	2.459	2.415	2.433	2.508	2.446	2.287	2.143
	Ebn at Noon		173	225	257	265	275	280	270	263	255	218	161	132
	Edh at Noon		14	21	27	33	33	32	33	30	24	19	13	10
All-Sky Solar Radiation	RadAvg		99	291	778	1235	1579	1792	1618	1275	852	381	127	67
	RadStd		16	53	91	143	148	112	187	139	110	68	29	7